ZeTheta Algorithms Private Limited

Full Stack Engineer Assessment — Final Project Submission

Project 1 B: Custom API Integration — SAP S/4HANA to Analytics (Course ID: 138518)

Candidate: Tejas H R | **Submission Date:** November 2026 | **Status:** 100% Verified & Passing

1. Executive Summary & Compliance Matrix

This submission provides the complete, production-grade implementation of the **Custom API Integration Gateway** connecting **SAP S/4HANA ERP Financials** to the ZeTheta Financial Analytics platform. The system implements strict Pydantic v2 data contracts across ERP modules (FI-GL, FI-AP, FI-AR, CO-OM), delta OData extraction with cursor pagination, automated field normalization with 100% mathematical double-entry reconciliation (Debit == Credit), enterprise resilience (Circuit Breaker, DLQ, exponential backoff), and OAuth 2.0 with HMAC-SHA256 signature verification.

Deliverable Part	Status	Technical Implementation
Part 1: Integration Architecture	**Completed**	Microservice bridge between S/4HANA and ZeTheta Analytics
Part 2: OpenAPI 3.1 & Data Contracts	**Completed**	OpenAPI 3.1 specification and Pydantic v2 schemas for ERP modules
Part 3: SAP Extraction Layer	**Completed**	OData v4 simulation with cursor pagination and delta sync
Part 4: Transformation & Reconciliation	**Completed**	Automated normalization with 100% Debit/Credit mathematical audit
Part 5: Resilience, Retries & DLQ	**Completed**	3-state Circuit Breaker, exponential backoff, and Dead-Letter Queue
Part 6: Security Protocols	**Completed**	OAuth 2.0 Bearer tokens and HMAC-SHA256 replay attack defense
Part 7: Full Stack Automated Testing	**Completed**	Automated test suite (`test_integration.py`) — **100% pass rate**
Final Part: Technical Documentation	**Completed**	Architecture specs, field mapping dictionaries, and runbooks

2. Architecture Overview

The integration pipeline follows an event-driven extract-transform-load (ETL) pattern:

- **Extraction:** Consumes SAP S/4HANA OData services (`A_JournalEntry`, `A_SupplierInvoice`, `A_CustomerInvoice`) supporting `$top`, `$skip`, and delta timestamps.
- **Resilience:** A 3-state Circuit Breaker (`CLOSED`, `OPEN`, `HALF_OPEN`) protects against cascading ERP timeouts. Poison pill payloads are isolated into an in-memory Dead-Letter Queue (DLQ).
- **Transformation:** Translates ERP posting keys (`SHKZG` 'S'/'H') into normalized double-entry debit/credit streams while verifying mathematical reconciliation.
- **Security:** Enforces OAuth 2.0 token validation and HMAC-SHA256 payload integrity signatures with a strict 300-second timestamp drift limit.

3. Automated Test Verification Results

The automated integration test suite (`test_integration.py`) was executed with 100% pass rate:

```
============================================================
Starting S/4HANA Custom API Integration Test Suite
============================================================
[PASS] Test 1: OpenAPI 3.1 specification conforms to enterprise standards.
[PASS] Test 2: SAP extraction layer extracted 85 records across paginated cursors.
```

```
[PASS] Test 3: 100% mathematical reconciliation verified. Debit=Credit=Rs. 1,672,500.00.
[PASS] Test 4: Fault tolerance engine (Circuit Breaker & DLQ) verified.
[PASS] Test 5: Security layer (OAuth 2.0 & HMAC-SHA256 replay defense) verified.
[PASS] Test 6: End-to-end FastAPI integration gateway verified.
============================================================
ALL 6 AUTOMATED INTEGRATION TESTS PASSED! (100% Pass Rate)
============================================================
```

4. Production File Structure

- `sap_models.py`: Pydantic v2 schemas for raw SAP documents and analytical models.
- `sap_extractor.py`: S/4HANA OData extraction and pagination engine.
- `transformer.py`: Field mapping, reconciliation audit, and working capital calculators.
- `resilience.py`: Circuit breaker, retry backoff, and dead-letter queue.
- `security.py`: OAuth 2.0 and HMAC-SHA256 signature verifier.
- `app.py`: FastAPI integration gateway endpoints.
- `test_integration.py`: Complete automated test suite.